Mohd Abdul Wasay Talha

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SUMMARY

Systems-focused software engineer specializing in TypeScript, Node.js, PostgreSQL, Redis, AWS, WebSockets, and distributed backend systems. Built production-grade applications involving event-driven architectures, real-time communication, background job processing, semantic search, and high-concurrency services.

ACHIEVEMENTS

BuildMyIdea - *Founder & Developer* Product Hunt Link

- Independently designed, built, and publicly launched BuildMyIdea, a startup–developer marketplace with live payment/escrow infrastructure, iterated the product twice based on direct user feedback and grew it to real, organically acquired users without funding, a team, or marketing spend.

PROJECTS

TradeX | *Bun, TypeScript, Redis Streams, PostgreSQL, WebSockets, Express* GitHub Link

- Architected a **highly concurrent spot trading exchange** using **Redis Streams** to **decouple distributed services** including the backend, matching engine, WebSocket server, WebSocket client, and background workers, enabling **asynchronous communication** and scalable order processing.
- Engineered an **in-memory matching engine** supporting order execution, partial fills, achieving a **90ms p95 latency**.
- Implemented a **zero-data-loss** disaster recovery system based on **Event Sourcing**, combining automated **Redis RDB snapshots** (backed up to AWS S3) with deterministic stream replay to restore volatile engine state after crashes.
- Developed a **real-time market data WebSocket server** that consumes asynchronous engine events to fan out orderbook depth updates to subscribers with **minimal latency**.
- Ensured durable, eventually consistent financial record-keeping by sinking executed trades into PostgreSQL via a dedicated Prisma consumer worker.

Build My Idea | *Express, PostgreSQL, Docker, Redis, BullMQ, pgvector, WebSockets* Explanation Link | Live Link

- Engineered a **full-stack monorepo platform** with seamless collaboration and team formation between early-stage startups and developers through scalable backend services and real-time communication.
- Launched **instant push notifications** via **Redis Pub/Sub**, broadcasting critical platform alerts.
- Improved transaction throughput by **30%** by eliminating $N + 1$ **queries** and integrating **bulk inserts** for payment processing and automated Razorpay payouts.
- Built a background worker to generate embeddings for projects, powering pgvector-based **semantic search** that matches projects by skill relevance rather than keywords. Offloading generation from the request path to keep the API responsive and **increased the matched outputs by 25%**.

Better Up Time - Monitoring Platform | *TypeScript, Express.js, Redis, AWS Lambda* Live Link

- Developed a full-stack **uptime monitoring platform** performing continuous website health checks, delivering real-time status dashboards and historical performance tracking that **reduced critical incidents by 20%**.
- Designed a **fault-tolerant monitoring architecture** with a **multi-region Redis-backed job queue** to coordinate globally distributed monitoring workers, enabling reliable **asynchronous task scheduling**, failure handling, and low-latency response-time collection.
- Deployed **serverless monitoring workers** on **AWS Lambda** to process asynchronous checks, **reducing end-to-end system latency by 35%** and maximizing polling throughput.

CERTIFICATIONS

AWS Certification May 2026 | Certificate Link

- Gained practical knowledge in cloud architecture fundamentals, setting up infrastructure, utilizing key AWS services (such as EC2, S3, and IAM).

EDUCATION

Mahatma Gandhi Institute of Technology Hyderabad

B.Tech in ECE (CGPA: 7.5), Minor Degree in AI & ML *2022 – 2026*

TECHNICAL SKILLS

Languages & Frameworks: JavaScript, Python, React, Next.js, Node.js, Express.js
Database & Real-time Communication: MongoDB, PostgreSQL, pgvector, WebSocket, Redis, BullMQ
DevOps & AI: Git, Linux, GitHub Actions, Docker, Kubernetes, AWS